Project 2: Soldering an LED Circuit

This project is an introduction to soldering and using soldering to make prototypes known as perfboards. Soldering is a way of **more permanently** attaching electrical components to a given circuit as opposed to **temporarily** attaching components using breadboards like in Project 1. Soldering allows you to be more versatile when building circuits and opens up new possibilities for circuits you design in the future.

Objectives

- Practice soldering.
- Practice circuit design.
- Light up the LED!

Task Description

Summary:

You will learn how to solder onto a PCB, which is more permanent in the design process of circuits than just a breadboard

In order to gain credit for this project, your circuit needs to work. You are absolutely free to go back and reference the previous lecture slides in order to ensure you are soldering correctly including placing your LED in the correct position.

What you will need:

- Soldering materials such as a soldering iron, solder and PCBs will be provided
- 1x LED
    - https://www.digikey.com/en/products/detail/qt-brightek-qtb/QBL8IW15C-CW2897/14664272
- 1x Pushbutton
    - https://www.digikey.com/en/products/detail/cui-devices/TS02-66-55-BK-160-LCR-D/15634374
- 1x 330 Ohm Resistor
    - https://www.digikey.com/en/products/detail/stackpole-electronics inc/CF14JT330R/1741399
- 1x battery and battery holder
- 1x PCB

Note: Soldered components will be permanently used and can not be reused for future circuits!

Checkpoints:

1. Conceptualize what the circuit will look like and which holes you will use for each component.
2. Put components into place in the PCB so that you know how each component will fit into the circuit.
3. Solder one joint at a time, while keeping leads of other components out of the way.
4. Solder your way to a full LED circuit, testing joints along the way to ensure full connectivity.

What to Submit

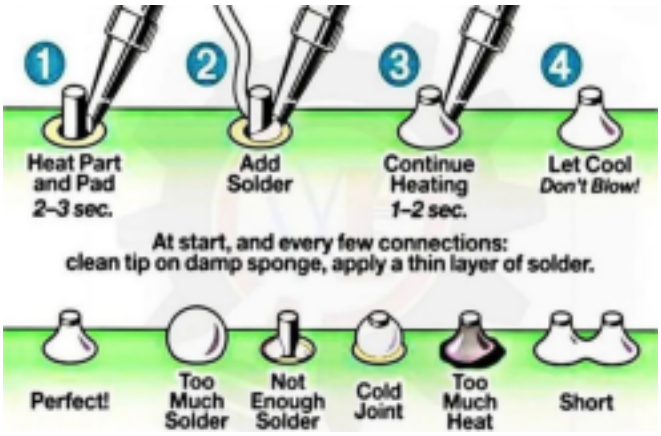
This project will be completed and demoed in lecture, so no submission is required.

Resources

L2 Github Folder:
https://github.com/IEEE-University-of-Florida/ops-resources-26-27/tree/main/L2%3A%20Soldering

Hints:

The below image shows the proper soldering technique and what the soldering joints should look like ideally. Joints that are not ideal tend to be due to one or more issues in technique as shown in the image.



Here are some general soldering safety precautions:

- Never touch the tip of the soldering iron. They are very hot (about 400°C) and will burn! The solder itself also gets hot. Never try to catch a hot soldering iron if you drop it! ● Always return the soldering iron to its stand when not in use. Never put it down on your workbench.
- Give any soldered surface a minute or two to cool down before you touch it. ● Ask for help! If you don't understand the technique, OPS Coordinators are here to guide you!

Sample Code: N/A
Schematics: N/A
Helpful Blogs:
Soldering overview
https://www.digikey.com/en/maker/tutorials/2022/how-to-solder